

FUSE TECHNOLOGY

This fuse technology guide will discuss basic fuse operating, application, and selection criteria concepts. The intended purpose of this section is to aid designers with the operation and characteristics of an overcurrent protection device and to assist in device selection.

Overcurrent fuses serve two main purposes:

- a. To protect components, equipment and people from risk of fire and shock caused by overcurrents.
- b. To isolate sub systems from the main system once a fault has occurred.

Overcurrents

Overcurrents exist when the normal load for a circuit is exceeded. It can either be an overload or short circuit condition.

An overload condition is any current flowing within the circuit path that is higher than the circuit's normal full load current. An overload is typically 2 to 5 times the magnitude of a circuit's normal operating current.

A short circuit is an overcurrent condition that leaves the normal current path and which greatly exceeds the normal full load current of the circuit by a factor of tens, hundreds, or thousands. Components and equipment can be damaged by both types of overcurrents.

Selecting Overcurrent Protection

During normal load conditions, the fuse must carry the normal operating current of the circuit without nuisance openings. However, when an overcurrent occurs the fuse must interrupt the overcurrent and withstand the voltage across the fuse after internal arcing. To properly select a fuse the following items must be considered:

- Voltage rating (ac or dc voltage)
- Current rating
- Normal operating current

- Ambient temperature
- Overload conditions and opening times
- Available short circuit current
- Melting Integral (I^2t)
- Pulse and In-rush characteristics
- Characteristics of equipment or components to be protected
- Physical size and available board space
- Standards requirements

Voltage Ratings

The voltage rating of the fuse must be greater than or equal to the maximum open circuit voltage. Because the fuse has such low resistance, the voltage rating becomes critical only when the fuse is trying to open. The fuse must be able to open quickly, extinguish the arc after the fuse element has melted and prevent the system's open-circuit voltage from re-striking across the open fuse element.

Current Ratings

The current rating of a fuse identifies its current-carrying capacity based on a controlled set of test conditions. Each fuse is marked with its current rating. This rating can be identified with a numeric, alpha, or color code mark. Marking codes can be found in each product's data sheet.

Normal Operating Current

The normal operating current of a circuit is the level of current drawn (in RMS or dc amperes) after it has been energized and is operating under normal conditions. An operating current of 80% or less of rated current is recommended for operation at 25°C to avoid nuisance openings. For example, a fuse with a current rating of 1A is usually not recommended in circuits with normal operating currents of more than 800mA. Further derating is required at elevated ambient temperatures.

Ambient Temperature

Ambient temperature is the temperature of the air immediately surrounding the fuse and is not necessarily room temperature. All electrical characteristics of a fuse are rated and validated at an ambient temperature of 25°C. Both higher and lower ambient temperatures will affect the fuse's opening and current carrying characteristics. This effect is demonstrated in temperature re-rating curves. Please refer to the re-rating curves for individual product series found in the Engineering Product Specifications located on the Cooper Electronic Technologies web site, or contact CET directly for technical assistance.

Overload Conditions and Opening Times

Specific attention must be given to first overload operating points. For fuses, the first overload point is usually between 200% to 300% of rated current. 400% is typically the first overload point for circuit protectors.

Breaking Capacity / Interrupting Rating

A fuse must be able to open the circuit under a short circuit condition without endangering its surroundings. The breaking capacity or interrupting rating of a protective device is the maximum available current, at rated voltage, that the device can safely open without rupturing. The breaking capacity or interrupting rating of a fuse must be equal to or greater than the available short circuit current of the circuit.

Melting Integral

The melting integral of a fuse, termed melting I^2t , is the thermal energy required to melt a specific fuse element. The construction, materials, and cross sectional area of the fuse element will determine this value. Each fuse series and ampere rating utilize different materials and element configurations, and therefore it is necessary to determine the I^2t value for each fuse. Tests to determine the I^2t of a fuse are

usually performed with a fault current of at least 10x the rated current with a time constant of less than 50 microseconds in a DC test circuit. High-speed oscilloscopes and integral programs are used to measure very accurate I^2t values.

The melting I^2t of a fuse is one of the values used to assist circuit designers when selecting and properly sizing a fuse in a specific application. It can be compared to the thermal energy created by transient surge currents in a circuit.

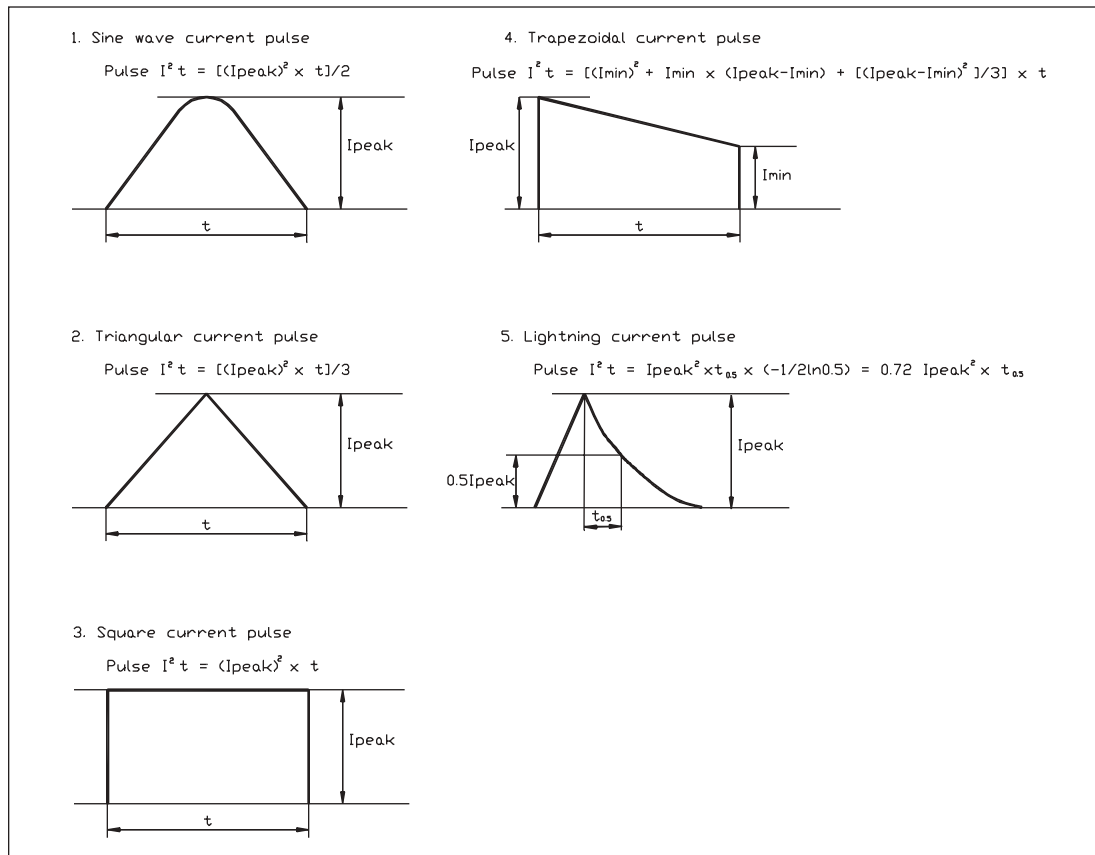
Surge and Pulse Current Characteristics

Transient surge or pulse currents are used to describe wave shapes that result from any startup, inrush, surge, or transient currents in a circuit. The pulse currents are normal for some applications. It is therefore important to size the fuse properly to allow these pulses to pass without nuisance openings or degradation of the fuse element. The fuse must then open within the limits specified by UL and CSA if the overload condition continues. The ability to resist surges is a function of the fuse design and/or classification relative to the surge pulse, duration frequency etc.

Pulse currents can produce thermal energy that may not be large enough to open the fuse but could possibly cause element fatigue and decrease the life of the fuse. To properly size a fuse and determine its surge withstand capability, the circuit's pulse energy should be determined and compared to the time current curve and I^2t rating of the fuse. The fuse's melting I^2t value must be greater than or equal to the pulse I^2t multiplied by a pulse factor.

The peak current and decay time define the pulse current characteristic or waveform. Pulses can generate different waveform shapes, which determines the formula used to calculate the pulse energy or I^2t . Refer to Table 1 to select the appropriate waveform and its corresponding pulse I^2t calculation.

Table 1. Pulse Waveshapes and I^2t Calculations



Fuse Surge Withstand Capability

The fuse's capability to withstand a surge pulse without causing thermal stress to the fuse element, which may result in nuisance openings, can be determined once the circuit's pulse I^2t is calculated. The circuit designer needs to properly size the fuse so that the fuse's melting I^2t value is greater than or equal to the pulse I^2t multiplied by a pulse factor F_p ($I^2t_{fuse} \geq I^2t_{pulse} \times F_p$).

The pulse factor is dependent on the construction of the fuse element. A wire-in-air constructed fuse element (ferrule fuses, 6125 and 1025 series for example) will be affected by the number and frequency of surge pulses the fuse is subjected to over the lifetime of the device. This construction design utilizes low-melting-point metals plated or deposited on the main element material to cause an "M" effect. If the fuse is sized improperly, low level pulse currents may cause the low-melting-point metals to alloy to the element without completely opening the element.

A series of pulse currents will eventually create enough heat to shift resistance or even permanently open the fuse. Thus it is important to take into account the number of pulse currents to which the fuse will be subjected.

Solid matrix fuses (for example 0603FA or 3216FF sized surface mount fuses) do not currently use an "M" effect for the element construction. The element will only then be affected by the thermal energy of each pulse, and will not normally degrade as a result of the number or frequency of pulses. Please refer to Table 2 to determine the pulse factor, F_p .

For example, a pulse current with an I^2t of 0.0823 and a pulse factor, $F_p=1.25$ would require the selection of a fuse to have a melting I^2t greater than or equal to 0.1029.

$$\text{Melting } I^2t_{fuse} \geq I^2t_{pulse} \times F_p$$

$$\text{Melting } I^2t_{fuse} \geq 0.0823 \times 1.25$$

$$\text{Melting } I^2t_{fuse} \geq 0.1029$$

It is important to note that the melting I^2t values of the fuse and pulse current that are compared must be calculated or tested at the same test conditions, most importantly the magnitude of the peak current must be the same. For example, if the pulse's peak current is 15A, then the fuse's melting I^2t must be calculated at 15A as well to fully understand its electrical characteristics at that magnitude of current. Please contact CET directly for technical assistance.

Table 2. Pulse Factor, F_p

Solid Matrix Construction

Number of Surge Pulses	Pulse Factor, F_p
1 to 100,000	1.25

Wire-in-Air Construction

Number of Surge Pulses	Pulse Factor, F_p
100	2.1
1,000	2.6
10,000	3.4
100,000	4.5

Time vs. Current Curves

A time current curve represents the relationship between a fuse's melting or clearing time and the magnitude of RMS or dc current. The characteristics represented on most published graphs usually indicate a fuse's average melting time when subjected to a certain level of current. The curves will typically demonstrate the ability to carry 100% of rated current, and then also represent the fuse's ability to open within the maximum opening time at designated overload points (typically 135% to 300% of the fuse rating). Time vs. current curves are a useful design aid for an engineer when specifying a fuse type or rating for an application. It is however recommended that fuse samples be tested in the actual application to verify performance.

Fuse Resistance

In most applications, the voltage drop across the fuse due to its internal and contact resistance is negligible. There are, however, certain critical applications where the fuse resistance must be considered and it is important that the circuit designer understands the fuse characteristics in order to select the proper fuse. Applications that are powered by low voltage batteries, typically 3V or less, and utilize fractional rated fuses with high resistance may require special attention be given to the voltage drop across the fuse.

Physical Sizes

There are numerous physical sizes of electronic fuses, including subminiature fuses. The most common ferrule designs are 5x15mm, 5x20mm and 6.3x32mm (1/4" x 1 1/4").

Subminiature fuses are often used when board space is limited. For applications of this type, there are through-hole and surface mount devices available. Standard package sizes for surface mount fuses are 0402 (1005), 0603 (1608), 1206 (3216), 6125, and 1025. These sizes are standard throughout the electronic industry. Through-hole axial and radial leaded products allow fuses to be PCB mounted. Standard ferrule fuses fitted with leads can also be mounted in this way.

Physical Sizes of Traditional Ferrule Fuses

5mmx20mm	.2" x .79"
1AG	1/4" x 5/8"
2AG (5mmx15mm)	.2" x .59"
3AG	1/4" x 1 1/4"
4AG	9/32" x 1 1/4"
5AG	13/32" x 1 1/2"
7AG	1/4" x 7/8"
8AG	1/4" x 1"

Standards

North American UL/CSA and IEC standards require significantly different time vs. current characteristics for overcurrent devices.

Typically the physical dimensions and materials used are similar; however, fuses built to different standards are not interchangeable because their element melting and opening times will differ when subjected to the same magnitude of current. It is therefore important for the circuit designer to consider that world standards may require different fuses.

Glossary of Terms

Ampere squared seconds I^2t

The melting, arcing, or clearing integral of a fuse, termed I^2t , is the thermal energy required to melt, arc, or clear a specific current. It can be expressed as melting I^2t , arcing I^2t or the sum of them, clearing I^2t .

Arcing time

The amount of time from the instant the fuse link has melted until the overcurrent is interrupted, or cleared.

Clearing time

The total time between the beginning of the overcurrent and the final opening of the circuit at rated voltage by an overcurrent protective device. Clearing time is the total of the melting time and the arcing time.

Fast acting fuse

A fuse which opens on overload and short circuits very quickly. This type of fuse is not designed to withstand temporary overload currents associated with some electrical loads. UL listed or recognized fast acting fuses would typically open within 5 seconds maximum when subjected to 200% to 250% of its rated current. IEC has two categories of fast acting fuses:

- F = quick acting, opens 10x rated current within 0.001 seconds to 0.01 seconds
- FF = very quick acting, opens 10x rated current in less than 0.001 seconds

Fuse

An overcurrent protective device with a fusible link that operates and permanently opens the circuit on an overcurrent condition.

Overcurrent

A condition which exists in an electrical circuit when the normal load current is exceeded. Overcurrents take on two separate characteristics-overloads and short circuits.

Overload

Can be classified as an overcurrent which exceeds the normal full load current of a circuit by 2 to 5 times its magnitude and stays within the normal current path.

Resistive load

An electrical load which is characterized by not drawing any significant inrush current. When a resistive load is energized, the current rises instantly to its steady state value, without first rising to a higher value.

RMS Current

The R.M.S. (root mean square) value of any periodic current is equal to the value of the direct current which, flowing through a resistance, produces the same heating effect in the resistance as the periodic current does.

Short circuit

An overcurrent that leaves the normal current path and greatly exceeds the normal full load current of the circuit by a factor of tens, hundreds, or thousands times.

Time delay fuse

A fuse with a built-in time delay that allows temporary and harmless inrush currents to pass without operating, but is so designed to open on sustained overloads and short circuits. UL listed or recognized time delay fuses typically open in 2 minutes maximum when subjected to 200% to 250% of rated current. IEC has two categories of time delay fuses:

- T = time lag, opens 10x rated current within 0.01 seconds to 0.1 seconds
- TT = long time lag, opens 10x rated current within 0.1 seconds to 1 second

Voltage rating

A maximum open circuit voltage in which a fuse can be used, yet safely interrupt an overcurrent. Exceeding the voltage rating of a fuse impairs its ability to clear an overload or short circuit safely.